

# Neiv Gupta

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## EDUCATION

### University of California, Los Angeles

Los Angeles, CA

Bachelor of Science, Computer Science

Expected June 2027

- **Relevant Coursework:** Algorithms & Complexity, Computer Systems Architecture, Software Construction, Electrical & Electronic Circuits, Operating Systems, Logic Design of Digital Systems, Discrete Structures

## EXPERIENCE

### Software Engineering Intern

Apr. 2025 – Sep. 2025

ThinkScan Technologies

Pleasanton, CA

- Developed an object detection and scene reasoning AI Agent for defense applications and field deployments.
- Optimized LLM using **RAG** pipeline over **FAISS** vector store for semantic retrieval across defense imagery.
- Implemented **Laplacian variance** and bilateral filtering to preprocess noisy defense imagery for inference.

### Computer Vision Researcher

June 2024 – Aug. 2024

Argonne National Laboratory

Lemont, IL

- Deployed **TensorFlow/PyTorch** CNN pipelines on SAGE edge nodes for real-time environmental monitoring.
- Built high-performance **CLIP**-based zero-shot classification achieving **93%** mAP across 12 environmental classes.
- Fine-tuned **OpenCLIP ViT-B/32** models on domain-specific environmental datasets for improved accuracy.

### Geographic Information Systems Intern

Apr. 2023 – Aug. 2024

Northern Change Research Laboratory, Brown University

Providence, RI

- Executed **ResNet-50** transfer learning with TensorFlow/PyTorch, processing Sentinel-2 multi-spectral imagery.
- Engineered data pipelines using GDAL/Rasterio, creating **15K+** labeled training samples via QGIS digitization.
- Quantified glacial ice loss contributing to **sea level rise** projections using **geodetic mass balance** calculations.

### Student Researcher

Nov. 2021 – Nov. 2023

Doer School of Sustainability, Stanford University

Palo Alto, CA

- Implemented **Random Forest** and **XGBoost** learning algorithms for Sierra Nevada tree species classification.
- Achieved **96%** classification accuracy using ensemble methods on **100,000+** labeled forest imagery samples.
- Developed **LSTM RNN** architecture for wildfire progression prediction from multi-spectral satellite imagery.

## PROJECTS

### CudaFire 🔥 | CUDA, C++17, C, CMake, GDAL, OpenGL

Dec. 2025 – Jan. 2026

- Built GPU-accelerated wildfire spread simulator using **CUDA** and the **Rothermel** fire behavior model.
- Processed **8.7 million terrain cells** in parallel using **8-connected cellular automaton** on RTX 3080.
- Optimized CUDA kernels with 16×16 thread blocks, achieving **7,643× real-time** simulation performance.
- Integrated GeoTIFF terrain ingestion via **GDAL** and real-time **OpenGL** 3D visualization pipeline.

### BruinMarket 🍌🍷 | Go, React, PostgreSQL, WebSockets, Docker, Railway, Vercel

Oct. 2025 – Nov. 2025

- Launched full-stack UCLA-exclusive student marketplace with real-time peer-to-peer transactions and messaging.
- Architected backend using **Go** with **PostgreSQL** database and **JWT** authentication and email verification.
- Deployed production app on **Railway** and **Vercel** with custom domain configuration and **CI/CD** pipeline.

### YUM 🍔 | React Native (Expo), Node.js, Express.js, MongoDB Atlas, REST APIs

Mar. 2025 – Jun. 2025

- Developed UCLA mobile dining app providing live dining updates, commenting workflows, and personal profiles.
- Architected full-stack **MERN** mobile app with **JWT**-based authentication and real-time state synchronization.
- Implemented **RESTful API** endpoints with Express.js middleware and **MongoDB** aggregation pipelines.

### Stairmasters 🦯 | Swift

Aug 2024 – Dec. 2024

- Developed Swift iOS accessibility app helping UCLA students with disabilities find accessible campus routes.
- Leveraged Apple's MapKit framework with **MKDirections API** for accessible route calculations and navigation.
- Mapped elevator access points using Swift Core Location framework for wheelchair-accessible campus navigation.

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C++, Go, JavaScript, TypeScript, HTML/CSS, XML, JSON, Swift, Bash, SQL, CUDA

**Frameworks:** PyTorch, TensorFlow, LangChain, React, React Native, Node.js, Express.js, Gin, Apple MLX

**Dev. Tools:** Git, Github, Docker, Railway, Vercel, REST APIs, WebSockets, MongoDB, PostgreSQL, OpenGL, FAISS

**Libraries:** NumPy, pandas, Matplotlib, Scikit-learn, OpenCV, CLIP, OpenCLIP, YOLO, XGBoost, Tailwind CSS